ABSTRACT

Customer churn refers to the situation in which a customer stops using a company's products or services. Predicting churn helps service providers identify customers who may leave and take suitable steps to improve customer retention. This project develops a machine learning classification model using the Random Forest algorithm to predict customer churn from the Telco Customer Churn dataset.

The notebook uses 7,043 customer records and 21 original columns. The data is cleaned and prepared by converting TotalCharges into a numeric field, replacing 11 missing numeric values with the median, removing the customerID field, and encoding categorical variables. The data is divided into training and testing sets in an 80:20 ratio. A Random Forest Classifier with 100 trees is trained and evaluated. The recorded test accuracy is 79.21%. Precision, recall, F1-score and feature importance are also examined.

1. INTRODUCTION

In subscription-based businesses such as telecommunications, retaining existing customers is important. A customer may discontinue a service because of cost, contract conditions, service quality or other reasons. When many customers leave, the company may lose revenue and may need to spend more to acquire new customers.

Machine learning can analyse historical customer information and identify patterns associated with churn. This project applies a Random Forest classification model to estimate whether a customer belongs to the churn or non-churn class. The model is trained using customer details such as tenure, monthly charges, total charges, contract type and service-related information.

2. PROBLEM STATEMENT

To build a machine learning model that analyses customer information and predicts whether a telecommunications customer is likely to leave the service (Churn = Yes) or continue using it (Churn = No).

3. OBJECTIVES

• To study the Telco Customer Churn dataset.

• To explore the dataset and identify missing or unsuitable values.

• To preprocess numerical and categorical customer information.

• To train a Random Forest Classifier using the prepared data.

• To evaluate the model using accuracy, precision, recall, F1-score and a confusion matrix.

• To identify the features that contribute most to the model's predictions.

4. SOFTWARE AND HARDWARE REQUIREMENTS

Software requirements

Python; Jupyter Notebook or Google Colab; NumPy; Pandas; Matplotlib; Seaborn; Scikit-learn; and Joblib.

Hardware requirements

A computer capable of running Python and a notebook environment. The notebook does not specify a minimum processor, RAM or storage configuration; ordinary student laptop hardware is generally sufficient for this dataset.

5. DATASET DESCRIPTION

The project uses the Telco Customer Churn dataset, identified in the notebook as the Kaggle dataset by blastchar. It contains 7,043 rows and 21 columns before preprocessing. The target column is Churn, which indicates whether a customer left the service.

Dataset item	Description
Number of records	7,043
Original columns	21
Target variable	Churn (Yes / No)
Churn = No	5,174 records
Churn = Yes	1,869 records
Predictor columns after preprocessing	19

The dataset includes customer and service details such as gender, senior-citizen status, partner and dependent information, tenure, phone and internet services, security and support services, contract, payment method, MonthlyCharges and TotalCharges.

6. METHODOLOGY

Step 1 – Import libraries

The notebook imports NumPy and Pandas for data handling, Matplotlib and Seaborn for visualisation, and Scikit-learn for splitting data, encoding labels, training the model and calculating evaluation metrics. Joblib is used to save the trained model.

Step 2 – Load the dataset

The CSV file WA_Fn-UseC_-Telco-Customer-Churn.csv is loaded into a Pandas DataFrame.

Step 3 – Explore the data

The notebook checks the dataset shape, data types, descriptive statistics, churn class counts and missing values.

Step 4 – Handle missing values

TotalCharges is converted from text to numeric. Blank values become missing numeric values; 11 such values are found and filled using the median. The customerID column is removed because it is an identifier rather than a useful prediction feature.

Step 5 – Encode categorical data

LabelEncoder is applied to object-type columns so that categorical values are represented numerically for the model.

Step 6 – Separate input and target

The Churn column is selected as the target variable (y), while the remaining 19 columns form the input features (X).

Step 7 – Split the data

The dataset is divided into 80% training data and 20% testing data. Stratification is used to preserve the class distribution, and random_state=42 makes the split reproducible.

Step 8 – Train and predict

A RandomForestClassifier with 100 trees and random_state=42 is fitted on the training data. The trained model predicts churn labels for the test data.

7. RANDOM FOREST ALGORITHM

Random Forest is a supervised machine learning algorithm used for classification and regression. It builds multiple decision trees using different samples and feature selections. For classification, the trees each produce a class prediction and the forest combines their votes to produce the final prediction.

Using multiple trees can make the prediction more stable than relying on a single decision tree. In this project, the Random Forest Classifier is configured with n_estimators=100, meaning that 100 decision trees are used.

General working process

• Prepare the training dataset.

• Build multiple decision trees using the training data.

• Obtain a prediction from each tree.

• Combine the tree predictions by majority voting.

• Return the final predicted churn class.

8. IMPLEMENTATION DETAILS

The notebook follows a step-by-step Python implementation. The main model configuration is RandomForestClassifier(n_estimators=100, random_state=42). The train_test_split function uses test_size=0.2 and stratify=y. After training, the model predicts the target labels for the test dataset.

The trained model is saved using Joblib with the filename random_forest_churn_model.pkl. This file can be loaded later for prediction without training the model again, provided the same preprocessing steps and feature order are applied.

9. RESULTS AND PERFORMANCE ANALYSIS

The model is evaluated on 1,409 test records. The following values are the actual metrics printed in the supplied notebook.

Evaluation metric	Recorded result
Accuracy	79.21%
Precision	63.73%
Recall	50.27%
F1-score	56.20%

Meaning of the metrics

Accuracy: The proportion of all test predictions that are correct. The recorded accuracy is approximately 79.21%.

Precision: Of the customers predicted as churners, the proportion who actually churned. The recorded precision is approximately 63.73%.

Recall: Of all actual churners, the proportion correctly identified by the model. The recorded recall is approximately 50.27%.

F1-score: The harmonic mean of precision and recall. The recorded F1-score is approximately 56.20%.

The notebook also generates a confusion matrix heatmap. The displayed notebook output does not include the numeric cell values of that matrix as text, so this report does not invent them.

The recall value indicates that the model does not identify all actual churners. Therefore, accuracy alone is not sufficient to judge the usefulness of this model, especially when the dataset contains more non-churn records than churn records.

10. FEATURE IMPORTANCE

Random Forest provides feature-importance values that indicate how much each input feature contributes to the model's decision process. The notebook reports the following five highest values:

Feature	Importance
TotalCharges	0.186841
MonthlyCharges	0.179192
tenure	0.154338
Contract	0.079631
PaymentMethod	0.050095

According to the model's feature-importance output, TotalCharges, MonthlyCharges and tenure are the three most important features. Contract and PaymentMethod are the next two highest among the listed features. Feature importance describes the model's use of features; it does not by itself prove that a feature causes churn.

11. ADVANTAGES AND LIMITATIONS

Advantages

• Can analyse multiple customer attributes together.

• Random Forest combines predictions from many decision trees.

• The notebook evaluates the model using several classification metrics.

• The trained model is saved for possible later use.

Limitations

• Recall for the churn class is approximately 50.27%, so many actual churners may not be identified.

• The dataset has more non-churn records than churn records.

• Label encoding is applied independently to categorical columns; the notebook does not save the encoders for later inference.

• The notebook does not report hyperparameter tuning or external validation.

12. FUTURE ENHANCEMENTS

The notebook suggests addressing class imbalance using class_weight='balanced' or a technique such as SMOTE, and performing hyperparameter tuning. Further improvements could include comparing the model with other classification algorithms, evaluating precision-recall trade-offs, and building a simple user interface for entering customer details and viewing a prediction. Any improved model should be tested on unseen data before use.

13. CONCLUSION

This project demonstrates the use of a Random Forest Classifier to predict customer churn using the Telco Customer Churn dataset. The notebook performs data exploration, converts TotalCharges into numeric form, fills 11 missing values with the median, removes the customer identifier, encodes categorical columns and splits the

data into training and testing sets. A Random Forest model containing 100 trees is trained and evaluated.

The model achieves 79.21% accuracy, 63.73% precision, 50.27% recall and a 56.20% F1-score on the test set. TotalCharges, MonthlyCharges and tenure are the most important features reported by the model. The results show that Random Forest can provide a useful starting point for churn prediction, while the recall score and class imbalance indicate that further improvement is needed.

14. REFERENCES

• Telco Customer Churn dataset by blastchar, Kaggle: https://www.kaggle.com/datasets/blastchar/telco-customer-churn

• Scikit-learn documentation – RandomForestClassifier: https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html

• Scikit-learn documentation – Model evaluation: https://scikit-learn.org/stable/modules/model_evaluation.html

• Pandas documentation: https://pandas.pydata.org/docs/

• Matplotlib documentation: https://matplotlib.org/stable/contents.html